

Appendix A

Critical Values and Confidence Intervals

Confidence Intervals

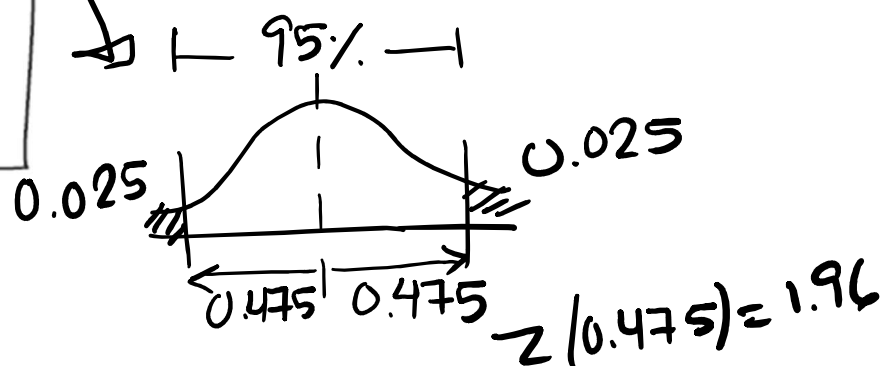
Type	Sample	Population	σ	Confidence Interval
mean	$n \geq 30$	any	known	$\bar{x} \pm z_c \frac{\sigma}{\sqrt{n}}$
mean	$n \geq 30$	any	unknown	$\bar{x} \pm z_c \frac{s}{\sqrt{n}}$
mean	$n < 30$	normal	known	$\bar{x} \pm z_c \frac{\sigma}{\sqrt{n}}$
mean	$n < 30$	normal	unknown	$\bar{x} \pm t_c \frac{s}{\sqrt{n}}$
proportion	$np \geq 5$ $nq \geq 5$	any		$p_s \pm z_c \sqrt{\frac{p_s(1-p_s)}{n}}$

Critical z-scores

Confidence Interval	Alpha	Tail	Critical z-score
0.99	0.01	two	± 2.57
	0.01	one	± 2.33
0.98	0.02	two	± 2.33
	0.02	one	± 2.05
0.95	0.05	two	± 1.96
	0.05	one	± 1.64
0.90	0.10	two	± 1.64
	0.10	one	± 1.28

Sample Size for Confidence Intervals

Type	Sample size
mean	$n = \left(\frac{z\sigma}{E} \right)^2$
proportion	$n = pq \left(\frac{z_c}{E} \right)^2$



Reference Table 2 Student's *t*-Distribution

Probabilities Under the <i>t</i> -Distribution Curve									
1-Tail	0.2000	0.1500	0.1000	0.0500	0.0250	0.0100	0.0050	0.0010	0.0005
2-Tail	0.4000	0.3000	0.2000	0.1000	0.0500	0.0200	0.0100	0.00200	0.0010
Conf Lev.	0.6000	0.7000	0.8000	0.9000	0.9500	0.9800	0.9900	0.9980	0.9990
df = $n - 1$									
1	1.376	1.963	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
50	0.849	1.047	1.299	1.676	2.009	2.403	2.678	3.261	3.496